Input, Output, and Processing for Humans

Activity	Input	Processing	Output
Speech	A thought	Form words	Spoken words
Reading	Written words	Understand meaning	Memory or response
Conversation	What you hear	Understand and decide	What you say

I/O and Processing for Computers

Program	Input	Processing	Output
Video game	Controller	Game logic	Screen and sound
Movie player	Network data	Decode video	Picture and audio
Console app	Keyboard values	Program logic	Printed text

Standard Input

- `cin` reads keyboard input and stores it in variables
- `>>` is the extraction operator
- Works with types such as `char`, `int`, and `float`

```
int age;
cin >> age;
```

Standard Output

- `cout` writes data to the console
- `<<` is the insertion operator
- Works with variables, literals, and constants

```
int age = 172;
cout << age;
```

Operators >>,<<

- Extraction and insertion operators can be chained

```
cin >> age >> name;
cout << age << name;
```

- `"\n"` starts a new line
- `endl` starts a new line and flushes the output stream

```
cout << "\n";
cout << endl;
```

PRACTICE

Lab

1. Define one `char`, one `int`, and one `float` variable.
2. Read all three values with one chained `cin` statement.
3. Print a label and value for each variable with `cout`.
4. Use both `"\n"` and `endl` in the output.

```
cin >> letter >> count >> price;
cout << letter << " " << count << " " << price;
```

PRACTICE

Assignment: Mad Libs Story

- Ask the user for two adjectives, two nouns, and one verb.
- Store each answer in a `string` variable.
- Insert the answers into a short story.
- Print the completed story with `cout`.

```
string adjective1, noun1, verb, adjective2, noun2;
```

PRACTICE

Mad Libs: Example Solution

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string adjective1, noun1, verb, adjective2, noun2;

    cout << "Adjective: "; cin >> adjective1;
    cout << "Noun: ";      cin >> noun1;
    cout << "Verb: ";      cin >> verb;
    cout << "Adjective: "; cin >> adjective2;
    cout << "Noun: ";      cin >> noun2;

    cout << "A " << adjective1 << " " << noun1
         << " loved to " << verb << ". Then it met a "
         << adjective2 << " " << noun2 << ".\n";
}
```

Boss Assignment

- Input the student name, subject, and marks out of 100.

Grade	C++ condition
A	`marks >= 90`
B	`marks >= 75 && marks < 90`
C	`marks >= 60 && marks < 75`
D	`marks < 60`

- Output the student's grade.
- **Challenge:** Process multiple students and count each grade.
- **Hint:** Use `if` / `else if`; the challenge also needs a loop.